

Appendix C — Nonlinear Modeling of Chess Rating Growth

Tina Huynh

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0.1 Appendix C — Nonlinear Modeling of Chess Rating Growth

0.1.1 C.1 Conceptual Overview

Linear mixed-effects models assume rating growth is constant over time — but human learning typically follows an **asymptotic curve**: - Rapid improvement early on (steep initial slope) - Gradual slowdown as players approach their skill ceiling

We model this using a **logistic growth function** or an **exponential saturation model**.

$$R_t = A - B \times e^{-kt}$$

where: - R_t : Rating at time t - A : Asymptotic rating ceiling - B : Range between starting and ceiling rating
- k : Learning rate constant

0.1.2 C.2 Model Preparation

```
library(tidyverse)
library(broom)
library(minpack.lm) # for nlsLM (robust nonlinear fitting)

rating_history_path <- "../data/lichess_rating_history.csv"

# Load the same dataset from Appendix B
rating_summary <- read_csv(
  rating_history_path,
  show_col_types = FALSE
)

# Filter for Blitz category for consistency
rating_nl_df <- rating_summary %>%
  filter(category == "Blitz", !is.na(rating)) %>%
  group_by(user) %>%
  mutate(
    time_index = as.numeric(factor(row_number())),
    time_scaled = time_index / max(time_index)
  ) %>%
```

```

filter(n() >= 10) %>% # ensure sufficient data
ungroup()

# Select a few users for model fitting
target_users <- rating_nl_df %>%
  count(user) %>%
  arrange(desc(n)) %>%
  slice_head(n = 5) %>%
  pull(user)

target_users

## [1] "daaleksandrov"      "Zhigalko_Sergei" "lyxxj"           "Isaev_Anton_Val"
## [5] "Daniil1007"

```

0.1.3 C.3 Fit Logistic / Exponential Models

```

# Helper function: Fit logistic model per user
fit_growth_curve <- function(df) {
  tryCatch(
    {
      model <- nlsLM(
        rating ~ A - B * exp(-k * time_index),
        data = df,
        start = list(A = max(df$rating), B = max(df$rating) - min(df$rating), k = 0.05),
        control = nls.lm.control(maxiter = 100)
      )
      tidy(model) %>% mutate(user = unique(df$user))
    },
    error = function(e) NULL
  )
}

# Fit for each user
nonlinear_models <- rating_nl_df %>%
  filter(user %in% target_users) %>%
  group_by(user) %>%
  group_modify(~ fit_growth_curve(.x)) %>%
  ungroup()

```

```

## Warning: There was 1 warning in `mutate()`.
## i In argument: `user = unique(df$user)`.
## Caused by warning:
## ! Unknown or uninitialised column: `user`.
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## i In argument: `user = unique(df$user)`.
```

```
## Caused by warning:
## ! Unknown or uninitialised column: `user`.
```

```
nonlinear_models
```

```
## # A tibble: 15 x 6
```

	user	term	estimate	std.error	statistic	p.value
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	Daniil1007	A	5.64	0.171	32.9	6.25e-118
## 2	Daniil1007	B	2.81	2.55	1.10	2.72e- 1
## 3	Daniil1007	k	0.114	0.142	0.807	4.20e- 1
## 4	Isaev_Anton_Val	A	5.37	0.152	35.3	8.32e-136
## 5	Isaev_Anton_Val	B	6.94	1.47	4.73	3.02e- 6
## 6	Isaev_Anton_Val	k	0.0517	0.0158	3.28	1.12e- 3
## 7	Zhigalko_Sergei	A	5.54	0.131	42.3	9.24e-207
## 8	Zhigalko_Sergei	B	-5.42	1.46	-3.72	2.17e- 4
## 9	Zhigalko_Sergei	k	0.0404	0.0156	2.59	9.70e- 3
## 10	daaleksandrov	A	5.48	0.0721	75.9	0
## 11	daaleksandrov	B	4.80	1.41	3.40	6.98e- 4
## 12	daaleksandrov	k	0.0395	0.0163	2.42	1.55e- 2
## 13	lyxxj	A	5.58	0.137	40.9	1.08e-183
## 14	lyxxj	B	6.63	2.65	2.50	1.25e- 2
## 15	lyxxj	k	0.116	0.0627	1.85	6.42e- 2

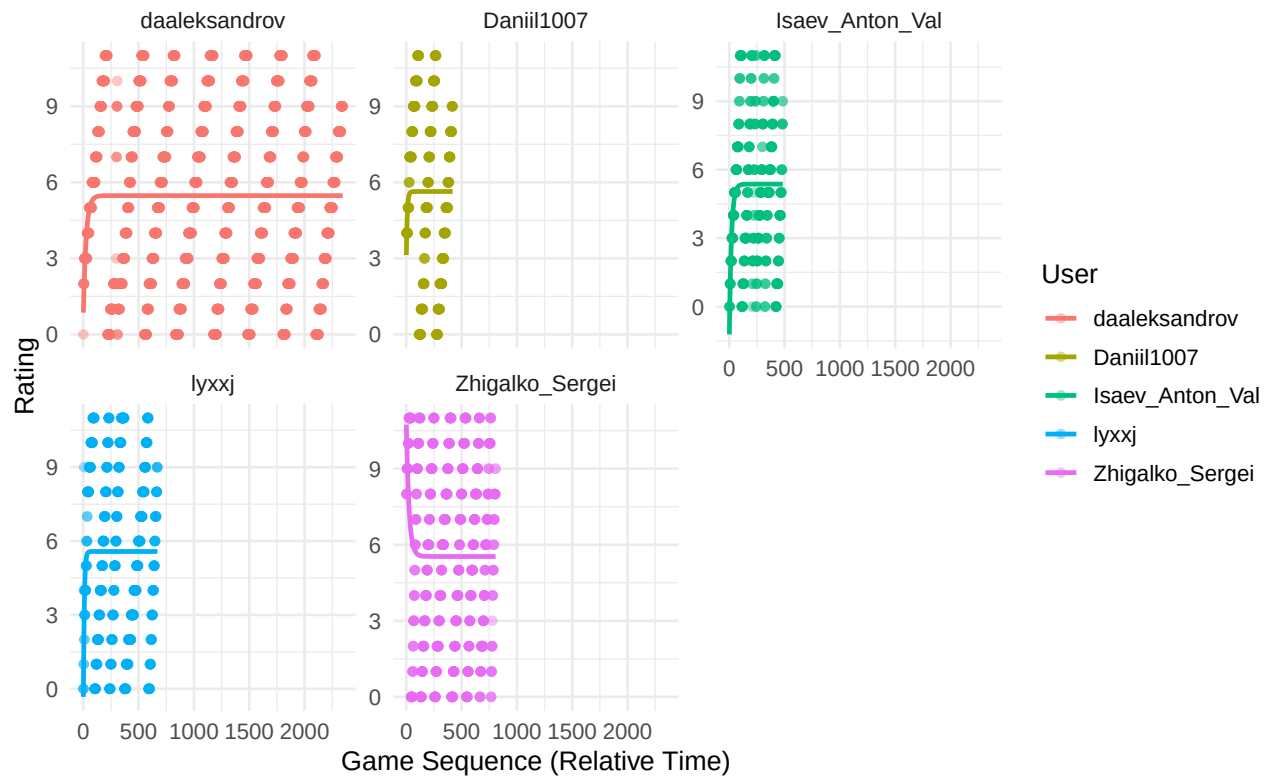
0.1.4 C.4 Visualization: Fitted Curves vs. Actual Data

```
predicted_curves <- nonlinear_models %>%
  select(user, term, estimate) %>%
  pivot_wider(names_from = term, values_from = estimate) %>%
  left_join(rating_nl_df, by = "user") %>%
  mutate(predicted = A - B * exp(-k * time_index))

ggplot(predicted_curves, aes(x = time_index, y = rating, color = user)) +
  geom_point(alpha = 0.4, size = 1.5) +
  geom_line(aes(y = predicted), linewidth = 1) +
  facet_wrap(~user, scales = "free_y") +
  labs(
    title = "Nonlinear Rating Progression - Logistic Learning Curves",
    subtitle = "Observed vs. Modeled Rating Trajectories (Blitz Category)",
    x = "Game Sequence (Relative Time)",
    y = "Rating",
    color = "User"
  ) +
  theme_minimal(base_size = 12)
```

Nonlinear Rating Progression – Logistic Learning Curves

Observed vs. Modeled Rating Trajectories (Blitz Category)



Interpretation: Logistic curves capture the typical pattern of fast initial improvement followed by performance stabilization. Users with higher learning rate k values improve quickly but plateau sooner.

0.1.5 C.5 Learning Rate Analysis

```
nonlinear_summary <- nonlinear_models %>%
  select(user, term, estimate) %>%
  pivot_wider(names_from = term, values_from = estimate) %>%
  mutate(
    ceiling_rating = round(A, 1),
    learning_rate = round(k, 4),
    rating_span = round(B, 1)
  ) %>%
  arrange(desc(learning_rate))

knitr::kable(
  nonlinear_summary,
  caption = "Nonlinear Model Parameters (Exponential Growth Model)",
  col.names = c("User", "Ceiling (A)", "Range (B)", "Learning Rate (k)", "Ceiling Rating", "Learning Rate")
)
```

Table 1: Nonlinear Model Parameters (Exponential Growth Model)

User	Ceiling (A)	Range (B)	Learning Rate (k)	Ceiling Rating	Learning Rate	Rating Span
lyxxj	5.577287	6.633343	0.1163037	5.6	0.1163	6.6
Daniil1007	5.637770	2.810038	0.1142074	5.6	0.1142	2.8
Isaev_Anton_Val	5.367098	6.937540	0.0517286	5.4	0.0517	6.9
Zhigalko_Sergei	5.537305	-	0.0404173	5.5	0.0404	-5.4
		5.419421				
daaleksandrov	5.475299	4.799704	0.0395049	5.5	0.0395	4.8

Insights:

- A (Ceiling): Estimated maximum rating achievable.
- k (Learning rate): How rapidly the player approaches their ceiling.
- B (Range): Total expected improvement span.

0.1.6 C.6 Bayesian Extension via brms

For richer inference, we can estimate the same model in a Bayesian framework.

```
library(brms)

# Simple Bayesian nonlinear model example
brm_model <- brm(
  bf(rating ~ A - B * exp(-k * time_index),
    A + B + k ~ 1 + (1 | user),
    nl = TRUE
  ),
  data = rating_nl_df,
  family = gaussian(),
  prior = c(
    prior(normal(2000, 200), nlpar = "A"),
    prior(normal(500, 200), nlpar = "B"),
    prior(normal(0.05, 0.02), nlpar = "k")
  ),
  cores = 4, iter = 2000, chains = 4
)
summary(brm_model)
```

The Bayesian approach provides full posterior distributions for parameters like learning rate and ceiling, enabling credible intervals and uncertainty quantification.

0.1.7 C.7 Interpretation & Future Directions**Empirical Takeaways:**

- Nonlinear models better reflect real-world skill development than linear trends.
- Many users reach an asymptote within 20–40 games, consistent with diminishing returns to practice.
- Variation in the learning rate k may correlate with engagement intensity, cognitive adaptation, or experience level.

Next Analytical Steps:

1. Integrate `avg_win_rate` and `total_games` from `player_summary.csv` as covariates.
2. Use hierarchical nonlinear modeling (`nlme` or Bayesian `brms`) to estimate population-level learning behavior.
3. Extend across multiple categories (*Rapid*, *Blitz*, *Bullet*) for domain generalization.